

Arachidonic-acid-derived eicosanoids: roles in biology and immunopathology.



Arachidonic acid (AA)-derived eicosanoids belong to a complex family of lipid mediators that regulate a wide variety of physiological responses and pathological processes. They are produced by various cell types through distinct enzymatic pathways and act on target cells via specific G-protein-coupled receptors. Although originally recognized for their capacity to elicit biological responses such as vascular homeostasis, protection of the gastric mucosa and platelet aggregation, eicosanoids are now understood to regulate immunopathological processes ranging from inflammatory responses to chronic tissue remodelling, cancer, asthma, rheumatoid arthritis and autoimmune disorders. Here, we review the major properties of eicosanoids and their expanding roles in biology and medicine.